Professorship of Energy Management Technologies
TUM School of Engineering and Design
Technical University of Munich

Energy Informatics Lab
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Assignment 1: Wind Power Forecasting

# Meeting	Location	Content
1st Session	N3815	Introduction of assignment; work on first dataset
Time until 2nd Meeting	Home	Work on assignment using first dataset
2nd Session	N3815	Work on final dataset; competition; presentations
Until next assignment	Home	Work on report

Background

This assignment concentrates on renewable energy forecasting, more precisely wind energy. The assignment will include a forecasting competition inspired by the Global Energy Forecasting Competitions held in 2012 and 2014 (GEFCom2012 and GEFCom2014) on kaggle.com and crowdanalytix.com. The data you will use in the assignment includes wind power data (normalized by the wind farm nominal capacity, to simplify things) for a wind farm in Western Denmark, Horns Rev with a nominal capacity of 160 MW, and weather forecast input information is from the European Centre for Medium-range Weather Forecasts (ECMWF - ecmwf.int), the world-leading research and operational weather forecasting center.

The work is organized in 2 stages. Stage 1 (training stage) concentrates on understanding the data and task and preparing for the competition. During this stage, students can get guidance and feedback from the course instructors. Stage 2 (competition stage) concentrates on the competition: students apply their forecasting method to a test dataset with unknown actuals. Forecasting results will be compared among the competing student groups. Grading is based on the rank in the competition, the interview being conducted in Session 2, and the report submitted before the beginning of the next assignment.

The aim of the assignment is to evaluate

- your understanding of the basics of renewable energy analytics,
- your ability to evaluate forecasts and have a critical eye on forecast quality,
- your ability to generate forecasts,
- your creativity in proposing approaches to forecasting.

The expected outcome of the assignment includes

- a report of maximum 2 pages (excluding appendices),
- your forecasts based on the test data, at regular intervals (1h).

Grading

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The following aspects of your work will be graded:

- Description of forecast strategies (background and motivations, type of methods and tools employed, process of making the forecast strategies evolve through the weeks, etc.)
- Rigor and creativity (originality and relevance of ideas being proposed, quality of formulation, etc.)
- Evaluation of forecasts and illustration (use of forecast verification metrics and tools, analysis of performance, deduction of potential ways to improve forecasts, etc.)
- Discussion (what went well, what went wrong, how the forecasts could be improved - in your opinion, etc.)
- Ranking (20 first, 10 second, 5 others, 0 if no submission)
- General presentation of the assignment report (flow of information, quality of English, quality of illustrations, etc.)

Task description

Wind power forecasts are essential inputs to several decision-making problems in electricity markets. It is therefore a good idea to understand how these forecasts can be generated. You will generate wind power forecasts for given periods, with weather forecasts as input, and with a long history of past realizations (2 years) to learn from. Such historical data includes weather forecasts themselves and the actual power generated by the wind farm. Therefore, what can be learned from these data is the relationship between weather forecasts and the power eventually produced at the wind farm. Based on this relationship learned from the past, your task is to apply it to input weather forecasts to generate wind power forecasts over the evaluation periods defined in the training and competition stage.

The weather forecasts to be used in the assignment consist of wind forecasts at 2 heights (10m and 100m above ground level). Wind forecasts are given in terms of their zonal and meridional components (u and v), which correspond to the projection of the wind vector on West-East and South-North axes, respectively. Weather forecasts are issued once a day at 00:00 and can have lead times between 1 and 24 hours ahead. The data and the various files provided for the assignment are described in detail in the Appendix.

Stage 1 – Training

In the training stage, students become familiar with the task, the data provided, and the expectations in terms of delivering the prediction outcome.

Historical data (weather forecasts and wind power measurements) will be provided for a period between 1.1.2015 and 1.2.2018. Based on these data, as well as input weather forecasts issued every day at midnight for the next 24 hours covering the period from 1.2.2018 to 1.3.2018, you are asked to predict the corresponding wind power generated in the time interval between 1.2.2018 and 1.3.2018 in hourly timesteps. The forecasts can be tested using the method evaluate_preds in the notebook forecast_evaluation.ipynb. The corresponding template file defining the format of the expected outcome file will be provided, as well. In the competition, the same evaluation script and file format will be used.

5 files will be provided at that stage:

- **TrainData_A.csv**: file containing past weather forecasts and wind power realizations for the period 1.1.2015 to 1.2.2018 to learn from.
- **WeatherForecastInput_A.csv**: file containing input weather forecasts for the period for which we aim to generate wind power forecasts (1.2.2018 to 1.3.2018).
- **ForecastTemplate_A.csv**: file providing the template for how the wind power predictions should be delivered.
- **Solution_A.csv**: file containing the actual wind power values measured at the wind farm for the period to be predicted.
- **forecast_evaluation.ipynb**: jupyter notebook containing a method for evaluating predictions.

Here are a few hints when one is to generate forecasts rapidly, and with limited knowledge/data:

- When recent power measurements are available, one can always use the last power measurement as the forecasts for any lead time in the future (so-called persistence).
- Some averaging of past measurements can also be a good idea, for instance over the last m hours.
- One can also think that wind power generation, for a given hour of the day, could be quite similar from day to day. Therefore, if aiming to issue a forecast for a certain hour of the day, the average of what was observed for that hour for the last m days can be a good first prediction.
- Expert rules can also work. E.g., you could set your basic rules such that: (rule 1) if predicted wind speed is less than 4 m/s, let's predict power to be 0; (rule 2) if predicted wind speed is greater than 12 m/s, let's predict power to be 1; (rule 3) Let's then draw a straight line between these two, to predict power generation as a linear function of predicted wind speed, for wind speeds between 4 and 12 m/s.

Stage 2 - Competition

In the competition stage, the student groups receive a final dataset to train on and are asked to generate predictions to compete against each other based on prediction quality.

Historical data (weather forecasts and wind power measurements) will be provided for a period from 1.1.2015 to 1.3.2018. Based on these data, as well as input weather forecasts issued every day at midnight for the next 24 hours covering the period from 1.3.2018 to 1.4.2018, you are asked to predict the corresponding wind power generated by the wind farm in the same time interval in hourly timesteps. The forecasts can be tested using the same method used above (evaluate_preds) in the notebook forecast_evaluation.ipynb. This method computes the *R2 score* of your predictions. For this metric, a naive model that always predicts the average power value would result in an R2 score of 0 and the best score you can achieve is 1. The corresponding template file defining the format of the expected outcome file will be provided, as well.

4 files will be provided at that stage:

- **TrainData_B.csv**: file containing past weather forecasts and wind power realizations for the period 1.1.2015 to 1.3.2018 to learn from.

- **WeatherForecastInput_B.csv**: file containing input weather forecasts for the period for which we aim to generate wind power forecasts (1.3.2018 to 1.4.2018).
- **ForecastTemplate_B.csv**: file providing the template for how the wind power predictions should be delivered.
- **forecast_evaluation.ipynb**: jupyter notebook containing a method for evaluating predictions. This is the same notebook used in the previous stage.

You are required to upload your predictions added to the file ForecastTemplate_B.csv to Moodle at a specified time.

Please name the uploaded file as follows:

Competition_1_Group_{Group_letter}.csv

The file will then be used by the tutors to generate a ranking of prediction qualities. The results will be presented at the end of the 2nd session.

Structure and contents of the report to be delivered

The report for the assignment should include:

- A description of the models and forecast strategies developed over time.
- An evaluation of the forecasts issued at stages 1. This may be based on, e.g., calculation of basic scores (bias, MAE, RMSE) overall and as a function of the lead time (between 1 and 24 hours ahead).
- A few plots of some of the predictions (i.e., for some days of your choice) issued at stage 1, along with corresponding power measurements, with a critical visual assessment of the predictions.
- Some proposals for how you would improve your prediction strategies if you were given more time and/or more data.
- The code as .ipynb or .py files, provided separately

The report and code must be uploaded as one .zip file to Moodle before the given due date.

Naming of Submissions

Each submission must be named according to the following pattern:

- Competition: f"Competition_1_Group_{Group_letter}.csv“
- Presentation: f"Presentation_1_Group_{Group_letter}.pdf“
- Report and Code: f"Report_Code_1_Group_{Group_letter}.zip“

As an example for the group submissions of Group E:

- “Competition_1_Group_E.csv”
- “Presentation_1_Group_E.pdf”
- "Report_Code_1_Group_E.zip“

Suggested readings

Morales JM, Conejo A, Madsen H, Pinson P, Zugno M (2014). Renewable energy sources - Modelling and forecasting. Chapter 2 in Integrating Renewables in Electricity Markets - Operational Problems, Springer Verlag: New York (PDF provided via Moodle) - also the references therein.

Appendix: Short description of data and various files

The files to be used for this assignment have all the necessary input weather forecasts, past wind power measurements, as well as templates for submitting the predictions, for the assignment. The various files include:

ForecastTemplate_A.csv

This file gives the template for submitting the predictions for 1.2.2018 to 1.3.2018. This file can be used as a basis for generating the predictions at Stage 1.

Variables (columns):

- TIMESTAMP: time stamps for the wind power prediction values to be generated
- FORECAST: prediction values themselves

ForecastTemplate_B.csv

This file gives the template for submitting the predictions for 1.3.2018 to 1.4.2018. This file must be used as a basis for submitting the predictions at Stage 2, i.e., the competition.

Variables (columns):

- TIMESTAMP: time stamps for the wind power prediction values to be generated
- FORECAST: prediction values themselves

Solution_A.csv

This file gives the true wind power measurements for 1.2.2018 to 1.3.2018, which will be used to calculate your scores and give you an idea of how good you did in the training stage.

Variables (columns):

- TIMESTAMP: time stamps for the wind power measurements, corresponding to the forecasts to be compiled in ForecastTemplate_A.csv
- POWER: measured power values themselves (normalized)

TrainData_A.csv

This file gives the set of data that can be used to find a relationship between weather forecast inputs (wind speed forecasts at 10m and 100m above ground level) and observed power generation. These data cover the period from 1.1.2015 to 1.2.2018. Weather forecasts are issued once a day at 00:00, while they have lead times between 1 and 24 hours ahead.

Variables (columns):

- TIMESTAMP: time stamps for the wind forecasts, and corresponding power measurements
- POWER: measured power values (normalized)
- U10: zonal component of the wind forecast (West-East projection) at 10m above ground level
- V10: meridional component of the wind forecast (South-North projection) at 10m above ground level
- U100: zonal component of the wind forecast (West-East projection) at 100m above ground level
- V100: meridional component of the wind forecast (South-North projection) at 100m above ground level

TrainData_B.csv

This file gives the set of data that can be used to find a relationship between weather forecasts inputs (wind speed forecasts at 10m and 100m above ground level) and observed power generation. These data cover the period from 1.1.2015 to 1.3.2018. Weather forecasts are issued once a day at 00:00, while they have lead times between 1 and 24 hours ahead.

Variables (columns):

- TIMESTAMP: time stamps for the wind forecasts, and corresponding power measurements
- POWER: measured power values (normalized)
- U10: zonal component of the wind forecast (West-East projection) at 10m above ground level
- V10: meridional component of the wind forecast (South-North projection) at 10m above ground level
- U100: zonal component of the wind forecast (West-East projection) at 100m above ground level
- V100: meridional component of the wind forecast (South-North projection) at 100m above ground level

WeatherForecastInput_A.csv

This file gives the set of input weather forecasts that can be used as input to predict wind power generation for 1.2.2018 to 1.3.2018, i.e., at Stage 1 of the assignment. These include wind speed forecasts at 10m and 100m above ground level. These weather forecasts are issued at 00:00 once a day, while they have lead times between 1 and 24 hours ahead.

Variables (columns):

- TIMESTAMP: time stamps for the wind forecasts
- U10: zonal component of the wind forecast (West-East projection) at 10m above ground level
- V10: meridional component of the wind forecast (South-North projection) at 10m above ground level
- U100: zonal component of the wind forecast (West-East projection) at 100m above ground level
- V100: meridional component of the wind forecast (South-North projection) at 100m above ground level

WeatherForecastInput_B.csv

This file gives the set of input weather forecasts that can be used as input to predict wind power generation for 1.3.2018 – 1.4.2018 at Stage 2 of the assignment. These include wind speed forecasts at 10m and 100m above ground level. Weather forecasts are issued once a day at 00:00, while they have lead times between 1 and 24 hours ahead.

Variables (columns):

- TIMESTAMP: time stamps for the wind forecasts
- U10: zonal component of the wind forecast (West-East projection) at 10m above ground level
- V10: meridional component of the wind forecast (South-North projection) at 10m above ground level
- U100: zonal component of the wind forecast (West-East projection) at 100m above ground level
- V100: meridional component of the wind forecast (South-North projection) at 100m above ground level